

Supplementary Material for Neural Space Time Memory

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A Additional Implementation Details

In this section, we provide comprehensive details regarding our training curriculum, optimization hyperparameters, baseline configurations, and the evaluation protocol for our generalized Novel View Synthesis experiments.

A.1 Two-Stage Training Curriculum

All stateful models are trained sequentially on short video clips consisting of $T = 4$ timesteps. At each timestep, the network processes 2 sparse input views to synthesize 1 target view. To optimize both the memory persistence and the free-viewpoint rendering flexibility, we employ a two-stage training curriculum, as follows.

Stage 1: Memory Bootstrapping (Pretraining). The objective of this initial phase is to explicitly encourage the network to build and rely upon its memory module. For the first frame of the sequence ($t = 0$), the input consists of one frontal view (cam00) and one back-facing view (cam08), providing broad geometric context. For subsequent frames ($t > 0$), the inputs transition to two random views. Crucially, the target supervision is a target view that is explicitly distinct from the $t > 0$ inputs. Note these target viewpoints remain *static* across the sequence. This rigid setup boosts memorizing the initial context to faithfully inpaint the targets at later timesteps.

Stage 2: Free-Viewpoint Adaptation (Finetuning). Once the memory module is bootstrapped, we adapt the network to handle completely unconstrained streaming scenarios. In this stage, the inputs at $t = 0$ are two random views, and the inputs at $t > 0$ are two random *other* views. Furthermore, the target viewpoints are no longer static; they are completely random, time-varying views that are strictly disjoint from the 4 input views. Ultimately, this allows for continuous free-viewpoint rendering from time-varying target views (e.g., a 360-degree spin around the subject) while continuously querying the generalized memory.

A.2 Training Hyperparameters

We implement our framework using JAX. Models are trained with a total batch size of 128 for 500,000 iterations during the pretraining stage, followed by an additional 25,000 iterations during the finetuning stage. We employ a cosine learning rate decay schedule with a linear warmup. During pretraining, the learning

rate is initialized at 1×10^{-6} , warms up over 4,000 steps to a peak of 4×10^{-4} , remains constant for 100,000 steps, and then decays down to a final value of 1×10^{-6} . During the finetuning stage, we retain the same schedule parameters but reduce the peak learning rate to 4×10^{-5} warming up over 4,000 steps and then remaining constant. Finally, for the image reconstruction objective across all models and stages, the perceptual loss weight is fixed at $\lambda_{\text{lpips}} = 0.5$.

A.3 Baseline Adaptations and Coordinate Alignments

To ensure a fair comparison, all baselines evaluated in Tab. 1 are trained on the exact same two-training curriculum. However, due to architectural differences, camera coordinate systems must be aligned differently:

LVSM [1]: We utilize the exact architecture described in the original LVSM paper. But for fair comparison, we use the same learning rate schedule as above and the same L2 and LPIPS loss. Because LVSM is a stateless, per-frame model, we construct its training batches by extracting only the final frame ($t = 3$) from our 4-timestep video clips, ensuring it trains on identical data distributions. For coordinate alignment, we relativize the camera coordinate system to each individual target camera over time (i.e., the specific target camera for that frame is set as the origin with an identity rotation).

LaCT [3] and NSTM (Ours): Stateful models must maintain coherent camera coordinate system across time and adapt to time-varying target cameras. Therefore, for LaCT and NSTM, we relativize all cameras in the sequence to the *first* target camera ($t = 0$), ensuring a stable, global coordinate space for the fast-weights to operate within.

Token-Mem [2]: Similar to LaCT, CUT3R requires a temporally consistent coordinate frame. We align all inputs to the average pose of the first frame’s input camera pair. We follow the original implementation’s stage 1 4-frame training hyperparameters including a state size of 768 tokens each with 768 feature channels. We train the model using cosine decay learning rate schedule with a peak LR of 10^{-4} for 300,000 steps. Differing from the paper, we optimize L2 and LPIPS losses consistent with the other methods compared.

B Memory Persistence Test

To provide a comprehensive view of baseline performance, we present an extended evaluation of our Memory Persistence Test. In Tab. 1, we evaluate the exact, unmodified *LVSM* [1] architecture, as described in Sec. A.3. For direct comparability, we benchmark this off-the-shelf stateless model against the Stage 1 checkpoints of our method and other stateful baselines.

The quantitative results in Tab. 1 align with the core conclusions established in the main paper. As a stateless architecture, LVSM fails to faithfully synthesize the occluded back view. NSTM significantly outperforms the LVSM baseline across all temporal horizons ($T = 4, 30, 60$), confirming the effectiveness of our memory mechanism.

Method	4 Timesteps					30 Timesteps					60 Timesteps				
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	mPSNR $\uparrow$	mSSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	mPSNR $\uparrow$	mSSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	mPSNR $\uparrow$	mSSIM $\uparrow$
LVSM	28.33	0.9647	0.0251	19.32	0.6179	28.14	0.9629	0.0265	18.95	0.5759	28.11	0.9628	0.0267	18.92	0.5737
LaCT-NVS	30.84	0.9746	0.0180	22.06	0.7348	25.67	0.9478	0.0369	16.94	0.5794	22.32	0.9275	0.0616	13.49	0.4942
Token-Mem	28.75	0.9638	0.0277	19.68	0.5955	28.66	0.9634	0.0284	19.42	0.5706	28.65	0.9634	0.0286	19.37	0.5674
Ours	30.71	0.9745	0.0186	21.78	0.7300	30.21	0.9712	0.0210	20.98	0.6646	30.09	0.9708	0.0212	20.82	0.6604

Table 1: Quantitative comparison. best , second best , and third best are colored. While LaCT-NVS performs the best in the short term (T=4), the standard update dot-product loss proves unstable for dynamic novel view synthesis resulting in a 39% drop in *mPSNR* over 60 timesteps. LVSM fails to reconstruct any view not directly seen in the inputs. Token-Mem also fails to recover occlusions from memorized state and maintains a constant performance across all timesteps. Please see comprehensive video comparison results in our supplementary website.

Method	Novel View Synthesis Over 60 Timesteps				
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	mPSNR $\uparrow$	mSSIM $\uparrow$
LVSM	29.47	0.9699	0.0224	20.28	0.6516
LaCT-NVS	19.41	0.9089	0.0913	10.05	0.3608
Token-Mem	27.49	0.9572	0.0312	18.07	0.5525
Ours	28.22	0.9660	0.0227	18.95	0.6597

Table 2: General Novel View Synthesis Comparison. We use two fixed frontal cameras as the input to NSTM, and use random target cameras at each time step to evaluate the novel view synthesis performance. best , second best , and third best are colored.

C General Novel View Synthesis Test

While the main paper focuses on the extreme “Memory Persistence Test” under forced occlusions, NSTM is fundamentally designed to be a highly versatile dynamic NVS framework. To evaluate its generalized synthesis capabilities, we conduct experiments under a standard, unconstrained streaming protocol. For this evaluation, we utilize the Stage 2 (Finetuned) checkpoints for all stateful models, as this stage specifically optimizes the network to handle time-varying target views without the rigid constraints of the initial memory bootstrapping phase (Sec. A).

Evaluation Protocol To simulate typical real-world tele-presence or broadcasting scenarios where input camera rigs are fixed, we restrict the streaming inputs to only two static frontal views (e.g., `cam15` and `cam00`). The model is then tasked with rendering the dynamic scene from completely random, time-varying target viewpoints. Because the target viewpoints are completely randomized at every frame, this unconstrained protocol inherently breaks the temporal continuity required for persistent memory accumulation. Note that this evaluation protocol is different from the training protocol, where the back view is always shown at the first frame. We utilize this setup strictly as a stress test for per-frame spatial generalization.

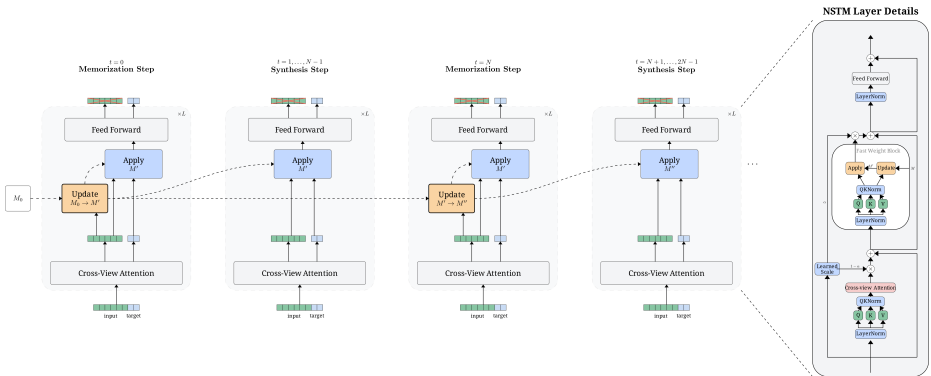


Fig. 1: Online Inference Protocol and NSTM Layer Details. Left: During inference, we run the memorization step periodically. We introduce the NSTM Layer consisting of cross-view attention, a fast-weight block and a feed-forward layer. We also incorporate a learned scale that allows the network to balance the contribution of memory and attention to each layer’s output.

Results and Discussions. Despite operating entirely out-of-distribution for a temporal memory model, as shown in Table 2, NSTM remains highly competitive in standard pixel-wise metrics (PSNR, SSIM) and perceptual fidelity (LPIPS), demonstrating that our auxiliary memory training does not compromise the model’s foundational spatial rendering capabilities.

D Online Inference Protocol

As shown in Fig. 1, NSTM explicitly decouples the frequencies of the Memorization Step and the Synthesis Step during online inference.

Asymmetric Processing Strides. Unlike models that rigidly couple memory updates and queries at every timestep, NSTM operates on independent execution strides. For our evaluation on the MVHumanNet++ benchmark, the sequences are aggressively subsampled, capturing only 1 frame every 25 frames from the raw 30 FPS videos. Consequently, the temporal gap between consecutive inputs is ~ 833 ms. Because of this large temporal spacing, we utilize a 1:1 memorization-to-synthesis ratio for these specific benchmark sequences. However, for real-world continuous streaming at 30 FPS, visual content is largely redundant and there is no need to perform heavy memory updates for every frame. Instead, we successfully amortize computational cost by employing a 29:1 ratio: The memory module executes the Memorization Step only once every 30 frames (i.e., every 1 second) to capture novel states, and we query this persistent memory at 30 FPS to execute the Synthesis Step for the target views.

E Detailed Model Architecture

Fig. 1 shows our NSTM layer in more details. Different from LaCT-NVS [3], we add an additional learned weight α to balance the contribution from the *cross-view attention* module and the *TTT* module.

F Foreground Matting and Background Augmentation

To facilitate clean isolation of the dynamic subject from the studio background, we output a 4-channel RGBA image. During training, we employ background augmentation: we composite both the ground-truth foreground and our model’s predicted RGBA output onto randomly sampled solid color backgrounds before computing the photometric loss. This augmentation forces the network to learn a highly accurate alpha matte. Consequently, at inference time, NSTM produces a clean, disentangled alpha channel, enabling seamless compositing of the synthesized human subject into arbitrary novel environments.

References

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